

Course Code: 20973

Workshop in Autonomous Systems Simulation

Course Booklet fall 2026

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Dear Student

Welcome to the Autonomous Systems Simulation Workshop. We are pleased to have you join us in this course, where you will explore simulation frameworks, develop autonomous system functionalities, and evaluate their performance in diverse scenarios.

This booklet provides you with essential information about the workshop, including the course schedule, activities, and project guidelines.

A dedicated course website is available, where you can find additional learning materials. The website serves as a communication platform for interacting with the teaching staff and fellow students. Additionally, you can access online learning resources and the university library services at the following links:

- Course Website:
- Library Services: <http://www.openu.ac.il/Library>
- Online Learning Platform: <http://telem.openu.ac.il>

If needed, you may schedule meetings via email at roiwe@openu.ac.il

For students studying abroad: Despite the physical distance, we strive to maintain close connections and provide support as needed. All essential course details are included in this booklet and on the course website. We highly recommend utilizing the online platform and all available learning resources.

Wishing you an enjoyable and productive learning experience!

Roi Wexler
Course Coordinator

Workshop Description

This workshop focuses on the simulation of autonomous systems, including autonomous vehicles, drones, and robots.

The primary goal is to familiarize students with an advanced simulation framework that enables the implementation of autonomous components, the addition of new functionalities, and the evaluation of these components' performance under diverse scenarios.

Each student will have the opportunity to choose a suitable simulation platform based on their interests and the guidance of their supervisor.

Simulation platforms available for use in this workshop include:

- [CARLA](#)
- [MuJoCo](#)

As a key part of the workshop, students will undertake a project—either individually or in pairs—where they will design, develop, and document an autonomous system simulation.

The project will require applying programming principles, defining simulation scenarios, and setting performance evaluation metrics.

Through this experience, students will develop technical proficiency in autonomous systems, problem-solving skills, and the ability to critically assess system behavior under diverse conditions.

Presentation:

Each student will prepare a presentation of approximately 30 minutes on his chosen project and present it to the rest of the students.

In addition, a copy of the presentation slides and a summary of the material must be submitted during the session.

presentation topics will be approved during the first three weeks of the semester.

Projects:

The workshop includes the development of a final project, utilizing the technologies learned in the course.

The project must adhere to programming principles.

Students may choose a topic from a list of suggested projects or propose their own ideas.

It is highly recommended working on this assignment in pairs. Working together will allow you to brainstorm ideas and tackle the tasks more efficiently. **If you need help finding a partner, try posting on the course forum or consult with your instructor.**

Should you choose to complete the assignment individually, the scope will be adjusted accordingly for a single person during the approval of the project plan.

Towards the mid of the semester, students will submit a project plan including the engineering solution (design) and present it to the class. A detailed explanation of this task can be found later in this manual.

The deadline for the final project submission is August 31st. By this date, students are required to submit their projects and schedule a 15–30 minute presentation meeting with the instructor.

Grading Policy:

- Project Plan Presentation: 35%
- Project Implementation: 65%

Code Quality	25%
SOLID Principles Implementation	15%
Reliability, Error Handling & System Robustness	15%
Project Structure & Design Implementation	15%
Documentation & Developer Experience	10%
Meeting KPIs and Performance Goals	20%
Self-Evaluation & Reflective Analysis	5% (bonus)

Course Materials:

- [CARLA Documentation](#)
- [MuJoCo Documentation](#)
- Additional readings and resources will be provided throughout the course.

Recommended Reads

- [S.O.L.I.D Principles](#)
- [The Clean Architecture - Clean Coder Blog](#)
- [Software Design Patterns & Architecture](#)
- [Kanban Guide for Software Development](#) will become relevant after the design solution is approved.

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Note; This is an English-language course that fulfills the credit requirements in English. This means that all course materials, presentations, and communication will be conducted in English.

Project Definition

In this workshop, students will develop a project involving the simulation of autonomous systems, such as self-driving vehicles, drones, or robots.

Students may choose a topic from the provided list of project suggestions or propose their own, subject to approval.

Projects could be submitted in pairs and must be completed according to the following schedule:

Topic selection

Three weeks into the semester, each student must select a project topic, get it approved by the instructor, and register for one of the scheduled meetings to present it to the class. Once approved, you will be expected to set up a Kanban board in Trello (link will be provided by the instructor) to manage your project's backlog, map out your development milestones, and populate it with detailed tickets that reflect your execution plan and technical tasks.

Project Presentations (design review) by mid-semester:

By mid-semester, students will present their project plan. The presentation will be focused on an explanation of the system design choices. Students must demonstrate a deep understanding of their project and be ready to answer related questions.

This session will simulate a professional design review meeting, where you will be expected to present the following:

- **Project Requirements** – Clear definition of functional and technical requirements, including constraints and expected outcomes.
- **System Architecture and Key Components** – Overview of the system structure solutions, main modules, and their interactions. You must include a clear architectural diagram and a sequence chart.
Design Alternatives: Present at least two valid architectural approaches. Compare them and justify your final choice by explaining the **trade-offs** and the **technical rationale** behind your decision.
- **Description of Simulation Scenarios** – Explanation of the real-world situations being simulated.
- **Performance Metrics** – Definition of key performance indicators (KPIs) for evaluating the system's efficiency and accuracy.
- **Challenges and Risks** – Identification of potential technical or conceptual challenges and proposed solutions.

- **Preliminary Prototype/Demo** – If possible, a basic implementation of core functionalities or a visualization of the system.
- **Evaluation Criteria** – How the system's success will be measured
- **Timeline** – Breakdown of individual contributions, milestones, and a work plan for completing the project.

Final Submission (by August 31st):

The project submission must include the following documentation:

- A design document outlining the architecture, key components, simulation scenarios, performance metrics, and all other elements from the Initial Planning Phase.
- A README file explaining how to install and run the system
- The code itself

Grading Criteria:

The project will account for 65% of the final grade, evaluated based on system design, stability, scenarios and tests plan, and bug resilience. The final grade also includes project documentation and presentation. A minimum final score of 60 is required to complete the course successfully.

project suggestions

Projects for CARLA (Autonomous Driving Simulation)

1. Intelligent Traffic Control System

Simulate an autonomous traffic management system that adapts in real time to congestion, optimizing traffic signals and rerouting vehicles for efficiency (project can focus on a part of this system).

2. Self-Driving Vehicle in Mixed Traffic

Develop a scenario where an autonomous car navigates through a mix of self-driving and human-driven vehicles, handling unpredictable behaviors safely.

3. Emergency Vehicle Navigation

Design an AI system that allows autonomous emergency vehicles (ambulances, fire trucks) to navigate through heavy traffic by coordinating with other self-driving emergency cars and the traffic around them, including traffic lights.

4. Smart Parking Assistant

Simulate an autonomous parking system where vehicles detect available spots and perform complex parking maneuvers in real-world conditions.

5. Weather-Adaptive Autonomous Driving

Test a self-driving car in various weather conditions, such as heavy rain, fog, or snow, and implement AI-based adjustments for visibility and traction control.

6. Safe Pedestrian Interaction

Develop an AI model that enables self-driving cars to predict pedestrian behavior, stop safely at crosswalks, and interact intelligently in dense urban environments (You can choose to focus on one component).

7. Cooperative Multi-Vehicle Navigation

Simulate a group of autonomous vehicles coordinating their movements at intersections and highways to improve traffic flow and reduce congestion (You can choose to focus on one component).

8. Reinforcement Learning for Lane Changing

Use reinforcement learning to train a self-driving car to execute safe and efficient lane changes on highways, considering traffic speed and density (You can choose to focus on one component).

9. Platooning of Autonomous Trucks

Develop a system where multiple autonomous trucks drive in close proximity to improve fuel efficiency and minimize aerodynamic drag (You can choose to focus on one component).

10. Smart Bus System for Public Transport

Simulate a network of autonomous buses that adjust routes based on passenger demand, traffic conditions, and real-time scheduling constraints (You can choose to focus on one component).

11. Simulation for an Autonomous Taxi System with On-Demand Dispatch

Develop a simulation of an autonomous taxi fleet management system, where the nearest available vehicle is dispatched based on passenger requests. The system will optimize route planning to ensure the fastest and most efficient journey to the destination (You can choose to focus on one component).

Projects for MuJoCo (Robotics and Physics-Based Simulation)

1. Quadruped Robot Terrain Adaptation

Simulate a four-legged robot (like a robotic dog) navigating rough terrain, learning to adjust its gait dynamically based on terrain feedback, or by implementing pretrained algorithm.

2. Dexterous Robotic Arm for Object Manipulation

Develop a robotic arm capable of precise manipulation tasks such as stacking, sorting, or assembling objects using reinforcement learning, or by implementing pretrained algorithm.

3. Human-Robot Collaboration in Industrial Tasks

Simulate an industrial environment where robots and human workers collaborate safely, focusing on motion prediction and adaptive responses (you can choose to focus on few pre-determined scenarios)

4. Balancing and Locomotion for Humanoid Robots

Implement a reinforcement learning model to enable a humanoid robot to maintain balance and walk across uneven surfaces.

5. Robotic Exoskeleton for Assisted Walking

Simulate a wearable robotic exoskeleton that adapts to a human's movement, providing support for rehabilitation or mobility enhancement.

6. Drone Precision Landing on Moving Platforms

Develop an autonomous drone that can land accurately on a moving platform, incorporating real-time sensor fusion and control algorithms.

7. Soft Robotic Gripper for Fragile Objects

Design a soft robotic gripper that can handle delicate objects like glassware or fruits without causing damage, optimizing grip force and surface adaptation.

8. Autonomous Warehouse Logistics

Develop a logistics system where multiple autonomous robots manage inventory, transport goods, and avoid collisions in a dynamic warehouse environment (You can choose to focus on one component).

9. AI-Controlled Robotic Soccer Players

Create a team of robotic soccer players trained using reinforcement learning to improve passing, goal-scoring, and teamwork strategies (You can choose to focus on one component like the goal keeper).